

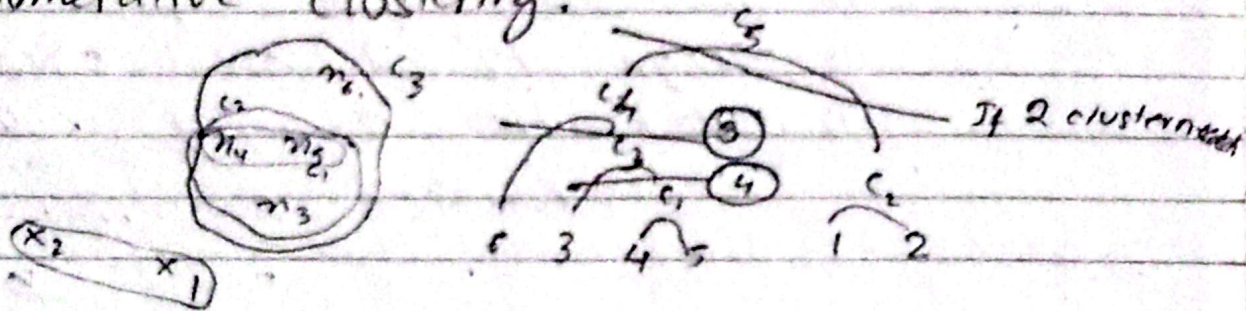
Agglomerative Hierarchical Clustering:

- In difficult pattern, K-Means Clustering fails.

Hierarchical Clustering

- Agglomerative Clustering
- Divisive Clustering

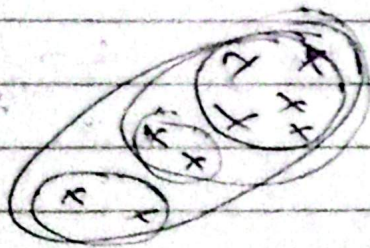
a) Agglomerative Clustering.



b) Divisive clustering.

opposite of agglomerative.

1 cluster $\rightarrow$ 2 clusters



breaking from 1 cluster $\rightarrow$ 2 clusters.

Algorithm.

Page No.

Date: / /

1) Initialize the Proximity Matrix:

	point1	P ₂	P ₃	P ₄
P ₁	0	-	-	-
P ₂	-	0	-	-
P ₃	-	-	0	-
P ₄	-	-	-	0

Diagram showing points P₁, P₂, P₃, P₄ and their distances. P₁ is at the top, P₂ is below it, P₃ is to the right of P₂, and P₄ is below P₃. Distances are indicated by 'x' marks and numbers: P₁ to P₂ is x₁, P₂ to P₃ is x₂, and P₃ to P₄ is x₂.

2) Make each point a cluster:

3) Inside a loop.

a) Merge the 2 closest cluster.

b) Update the Proximity Matrix.

c) combine two points first (P₃ & P₄) ^{closest on.}

	P ₁	P ₂	C ₁	0
P ₁	0	-	-	-
P ₂	-	0	-	-
C ₁	-	-	0	-

Do like this combine: C₁ → [P₃ & P₄], C₂ → [C₁ & P₂]
C₃ → [P₁ & C₂]

4) Until only one cluster left.

Types of Agglomerative Clustering.

1) Min (Single-link)

- can't distance. ~~be with~~ combine based on minimum distance

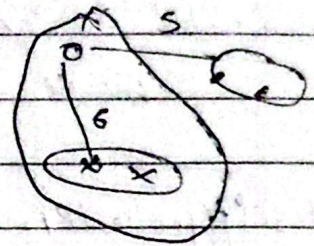
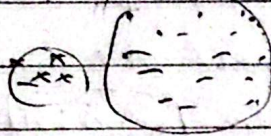
- Use if the distance betⁿ two cluster is fine.

- If outlier or little mixed the avoid this. ^{don't do}

2) Complete link (max)

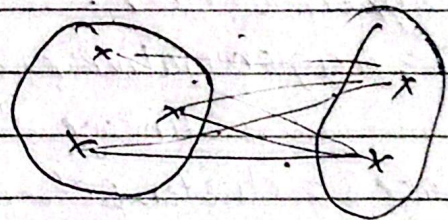
- based on max distance i
- better for outliers
- not use in ~~clust~~
- non-uniform cluster

[one higher another size low].



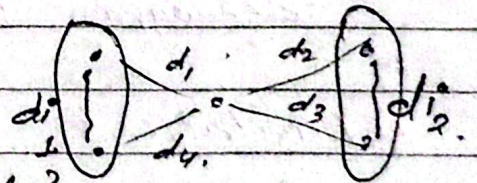
3) Group Average:

- calc all distance with each points of diff cluster.
- $\frac{d_1 + d_2 + d_3 + d_4 + d_5}{\text{total no. of points (5)}}$



4) Ward [default].

- calc centroid



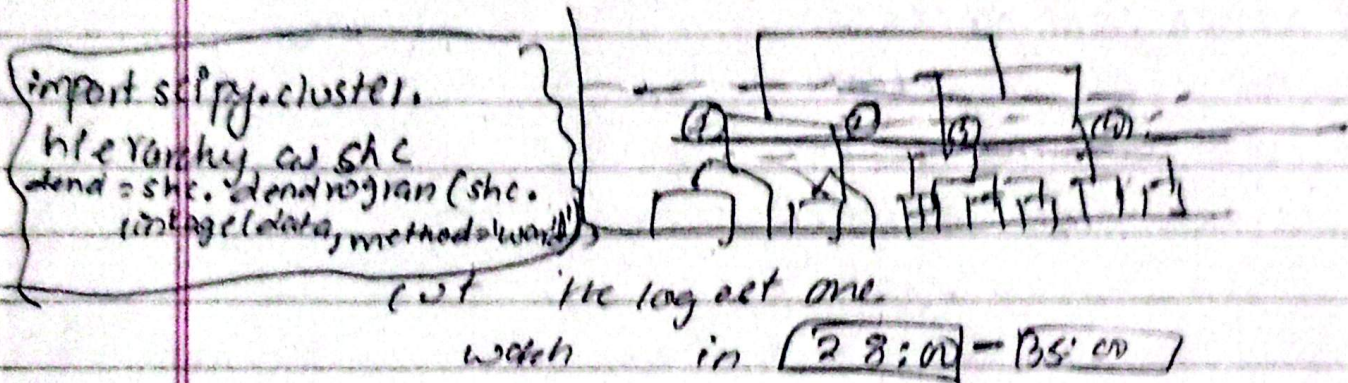
$$\text{distance} : d_1^2 + d_2^2 + d_3^2 + d_4^2 - d_{i_1}^2 - d_{i_2}^2$$

1A → 1B

- minimize the variance.

How to find ideal number of clusters.

make ~~cluster~~ dendrograms and cut the top one.



Hyperparameter.

cluster = Agglomerative clustering (n_clusters=2, affinity='euclidean',
linkage='ward', distance_threshold=None),
labels = cluster.fit_predict(data)

Benefits:

- dendrogram We can know hierarchy.

Limitations: - not in higher dataset.

For we have to find distance from each point to each other,
need to store all data in RAM, can't handle by not RAM.